EE659: A 0.2-3 GHz Inductor-Less LNA Using Noise-Canceling and Dual-Resistor Feedback Technique

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OBJECTIVE: Design an inductor-less LNA that achieves high gain, low noise figure, and proper input matching simultaneously by using noise-canceling and dual-resistor feedback techniques, and to verify its performance through simulation and practical implementation.

LNA - Low Noise Amplifier

1 Introduction

The rapid growth of 5G wireless communication systems has increased the demand for high-performance RF receiver circuits. In RF receivers, the Low Noise Amplifier (LNA) is a crucial component placed at the front-end to amplify weak signals received from the antenna while introducing minimal additional noise. The performance of the LNA significantly affects the overall sensitivity and signal quality of the receiver.

However, conventional LNA designs face challenges due to the trade-off between gain, noise figure, and input matching. Moreover, many traditional LNA architectures rely on inductors for impedance matching, which occupy a large chip area in CMOS technology. To overcome these limitations, this project focuses on the design of an inductor-less LNA using noise-cancelling and dual-resistor feedback techniques. The proposed design aims to achieve wide bandwidth, low noise figure, high gain, and compact chip area suitable for modern wireless communication applications.

- Amplify very weak signals coming from the antenna
- Add very little noise while amplifying the signal
- Maintain proper impedance matching (50 Ω)

2 Literature Review

2.1 Tradition LNA

Traditional LNA architectures such as resistor-feedback LNAs have been widely used in RF receiver front-ends. These designs aim to achieve low noise figure, high gain, and proper impedance matching. However, conventional LNAs often rely on on-chip inductors that occupy large chip area and suffer from trade-offs between gain, noise figure, and input matching.

I.Circuit Diagram

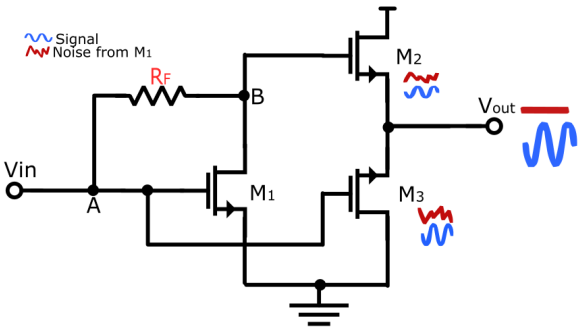


Figure 1: Schematic of the Traditional LNA

R_{in} calculation of Tradition LNA

Fig. 1 presents the conventional single-resistor feedback LNA. The noise-cancellation (NC) stage is constructed using transistors M_2 and M_3. The input resistance R_{in} is defined as

$$\frac{V_{in} - V_B}{R_F} = I_{in}$$

$$\frac{V_B - V_{in}}{R_F} + \frac{V_{in} - 0}{r_{ds1}} + g_{m1}V_{in} = 0$$

Now,

$$V_B = V_{in} - I_{in}R_F$$

Putting equation (3) into equation (2):

$$-I_{in} + \frac{V_{in} - I_{in}R_F}{r_{ds1}} + g_{m1}V_{in} = 0$$

$$\frac{V_{in}}{r_{ds1}} + g_{m1}V_{in} = I_{in} + \frac{I_{in}R_F}{r_{ds1}}$$

$$V_{in}\left(\frac{1}{r_{ds1}} + g_{m1}\right) = I_{in}\left(\frac{r_{ds1} + R_F}{r_{ds1}}\right)$$

Therefore,

$$R_{in} = \frac{V_{in}}{I_{in}} = \frac{r_{ds1} + R_F}{1 + g_{m1}r_{ds1}} \tag{1}$$

where g_{m1} and r_{ds1} are the transconductance and drain-source resistance of M_1, respectively. The noise generated by transistor M_1 is modeled as a current noise source. Two in-phase noise voltages are produced by $I_{n,M1}$ at node A ($I_{n,M1}R_S$) and node B ($I_{n,M1}(R_S + R_F)$). Transistor M_3 amplifies and reverses the polarity of the former noise voltage, while the latter noise passes through the buffer transistor M_2. If the following condition is satisfied, the noise voltages cancel each other at the output due to opposite phases: Here, $I_{n,M1}$ is noise current of M1. So, let

$$R_S = \text{input impedance at source}$$
$$R_F = \text{feedback resistance}$$

So, I_n flows through $(R_S + R_F)$.
At node A,

$$V_{A,n,M1} = I_{n,M1} \cdot R_S$$

At node B,

$$V_{B,n,M2} = I_{n,M1} \cdot R_F$$

M3 amplifies and reverses, whereas M2 passes noise (acts as buffer).
For zero noise condition:

$$-g_{m3}R_S + g_{m2}(R_S + R_F) = 0$$

$$\frac{g_{m2}}{g_{m3}} = 1 + \frac{R_F}{R_S}$$

Noise figure:

$$F = 1 + \frac{R_S}{R_F} + \frac{\gamma/\alpha}{g_{m2}}\left(\frac{1}{R_S} + \frac{3}{R_F} + \frac{2R_S}{R_F^2}\right) \tag{2}$$

Gain calculation

For M1,

$$\begin{aligned} i_{d1} &= g_{m1}v_{gs1} \\ v_{gs1} &= V_A \\ i_{d1} &= g_{m1}v_{in} \\ Voltage at output due to M1: & \end{aligned}$$

$$V_{out1} = -g_{m1}R_F v_{in}$$

Through noise cancelling transistor,

$$\begin{aligned} V_{out2} &= -\frac{g_{m2}}{g_{m3}}v_{in} \\ V_{out3} &= v_{in} \end{aligned}$$

$$\begin{aligned} V_{out} &= v_{in} - g_{m1}R_F v_{in} - \frac{g_{m2}}{g_{m3}}v_{in} \\ &= v_{in}\left(1 - g_{m1}R_F - \frac{g_{m2}}{g_{m3}}\right) \end{aligned}$$

$$\frac{V_{out}}{v_{in}} = 1 - g_{m1}R_F - \frac{g_{m2}}{g_{m3}}$$

Voltage Gain and Comparison

$$A_v = 1 - g_{m1}R_F - \frac{g_{m2}}{g_{m3}}$$

Under noise cancelling condition,

$$\frac{g_{m2}}{g_{m3}} = 1 + \frac{R_F}{R_S}$$

$$\begin{aligned} A_v &= 1 - g_{m1}R_F - \left(1 + \frac{R_F}{R_S}\right) \\ &= -R_F\left(g_{m1} + \frac{1}{R_S}\right) \end{aligned}$$

$$\boxed{A_v = -R_F\left(g_{m1} + \frac{1}{R_S}\right)}$$

Hence,

$$(A_v)_{\text{traditional}} = -R_F\left(g_{m1} + \frac{1}{R_S}\right) \tag{3}$$

$$(F)_{\text{traditional}} = 1 + \frac{R_S}{R_F} + \frac{\gamma/\alpha}{g_{m2}}\left(\frac{1}{R_S} + \frac{3}{R_F} + \frac{2R_S}{R_F^2}\right) \tag{4}$$

$$(R_{in})_{\text{traditional}} = \frac{r_{ds1} + R_F}{g_{m1}r_{ds1} + 1} \tag{5}$$

By comparing these equations:

Increasing R_F improves voltage gain and noise figure, but degrades impedance matching by increasing R_{in}.

Matching occurs when:

$$R_F \approx r_{ds1}$$

These architectures are commonly used in RF receiver front-ends to amplify weak signals received from the antenna. However, they suffer from several performance trade-offs, particularly between gain, noise figure, and input impedance matching.

Trade-off Between Noise, Gain, and Matching in Traditional LNA

The main limitation of traditional LNAs arises from the single-resistor feedback structure.

Role of Feedback Resistor

In a traditional feedback LNA, a resistor R_F is connected between the output and input of the amplifier. This feedback provides:

- Input impedance matching
- Broadband operation
- Improved stability

However, it also introduces performance trade-offs.

Gain vs Input Matching Trade-off

In a traditional feedback LNA:

- If the feedback resistor increases:
  - Gain increases
  - Noise performance improves
  - But input matching becomes worse
- If the feedback resistor decreases:
  - Input matching improves
  - But gain decreases

Therefore, it becomes difficult for the designer to optimize gain and input matching simultaneously. This occurs because the input resistance depends on both the transistor parameters and the feedback resistor.

Noise vs Gain Trade-off

Noise in an LNA mainly originates from:

- MOS transistor channel noise
- Thermal noise of resistors

In traditional LNAs, increasing the feedback resistor improves gain but also increases the noise contribution. As a result, achieving both low noise and high gain simultaneously becomes difficult. Additionally, the noise generated by the input transistor cannot be cancelled, which limits the minimum achievable noise figure (NF).

Area Problem in Traditional LNAs

Most conventional LNAs utilize inductors for:

- Input matching
- Bandwidth control
- Gain enhancement

However, inductors occupy a large chip area and reduce the advantage of CMOS integration. Therefore, traditional LNA designs are not ideal for compact RF systems.

Advantages of the Proposed LNA

To overcome the limitations of traditional LNAs, the proposed design incorporates:

- Noise cancellation technique
- Dual-resistor feedback
- Inductor-less architecture

These techniques enable the LNA to achieve:

- Low noise figure
- High gain
- Good input matching
- Small chip area
- Wide bandwidth

simultaneously.

2.2 Proposed LNA

The proposed LNA introduces an efficient architecture that integrates noise cancellation with dual-resistor feedback, enabling simultaneous optimization of critical RF parameters. This makes it highly suitable for broadband wireless communication systems, especially in sub-6 GHz applications.

A. Noise-Canceling Technique

I. Circuit Daigram

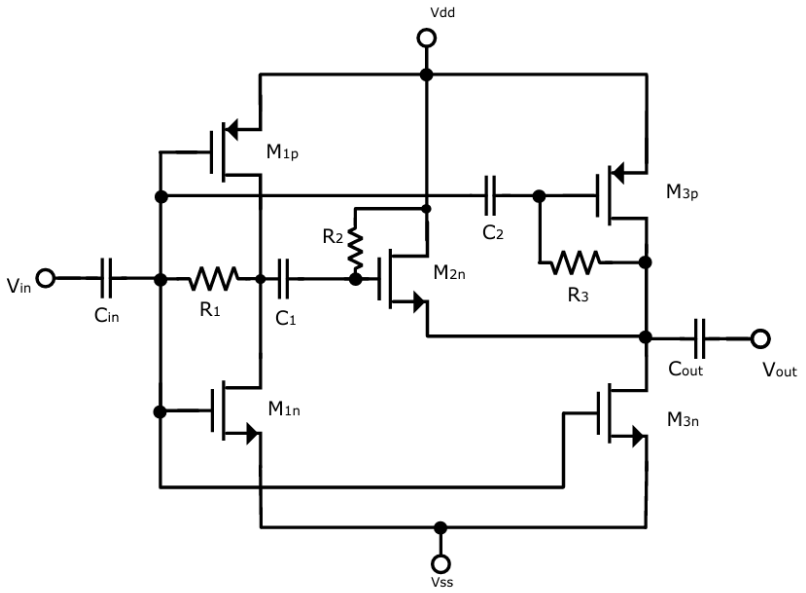


Figure 2: Schematic of the proposed LNA.

Fig. 2 illustrates the noise-cancellation path of the proposed LNA. Since the resistance R_2 is very small, its noise contribution is negligible. Therefore, the overall noise factor of the LNA can be expressed as:

$$F = 1 + F_{M1} + F_{M2} + F_{M3} + F_{R1} + F_{R3} \tag{6}$$

The first term represents the noise contribution from the source resistance, while F_i denotes the noise contribution of each circuit component. The thermal noise current of transistor $M1$ is given by:

$$I_{n,M1}^2 = 4k_BT\frac{\gamma}{\alpha}g_{m1} \tag{7}$$

where T is the absolute temperature and k_B is Boltzmann's constant. Due to the high gain of the input stage, the noise contributions from transistors $M2$ and $M3$ are negligible. The noise factor of transistor $M1$ can be written as:

$$F_{M1} = \frac{V_{n,\text{out}|M1}^2}{4k_BTR_SA_V^2} = \frac{I_{n,\text{out}|M1}^2}{4k_BTR_SG_m^2}$$

Further simplification leads to:

$$F_{M1} = \frac{\gamma}{\alpha}\frac{g_{m1}}{R_S}\left(\frac{\frac{1}{R_3} - g_{m3} + g_{m2}\left(\frac{R_3}{R_S} + \frac{R_1}{R_3} + 1\right)}{g_{m1} + \frac{1}{R_S} + \frac{1}{R_3}}\right) \tag{8}$$

Noise cancellation occurs when the numerator of (8) becomes zero, resulting in the condition:

$$\frac{g_{m3} - \frac{1}{R_3}}{g_{m2}} = 1 + \frac{R_3}{R_S} + \frac{R_1}{R_3} \tag{9}$$

- Compared to the conventional case, the term $\frac{1}{R_3}$ reduces the effective transconductance of M_2, while the ratio of noise voltages increases due to $\frac{R_1}{R_3}$, requiring a higher g_{m3} to satisfy the noise-cancellation condition.
- Despite the need for higher g_{m3}, the additional power consumption remains minimal due to current reuse and the high resistance of R_3.
- The proposed noise-canceling structure effectively suppresses the dominant noise from M_1, significantly reducing the overall noise figure, while the contributions from M_2, M_3, and R_3 remain negligible.

B. Dual-Resistor Feedback Technique

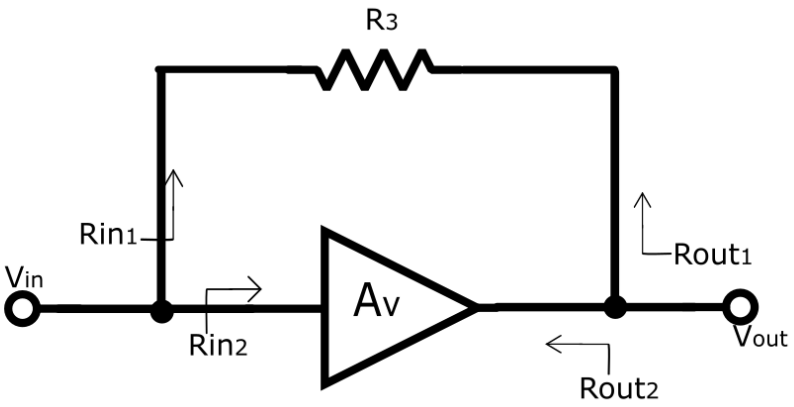


Figure 3: Block Diagram of the proposed LNA.

G_m Calculation

$$G_m = \frac{I_{out}}{V_{in}}$$

$$V_x = V_{in}(1 - g_{m1}R_1)$$

$$I_{M2} = g_{m2}V_x$$

$$I_{R3} = \frac{V_x}{R_3}$$

$$I_{mid} = \left(g_{m2} - \frac{1}{R_3}\right) V_x$$

$$I_{out} = g_{m3}V_x - I_{mid}$$

$$I_{out} = g_{m3}V_x - \left(g_{m2} - \frac{1}{R_3}\right) V_x$$

$$I_{out} = \left[g_{m3} - \left(g_{m2} - \frac{1}{R_3}\right)\right] V_x$$

$$I_{out} = \left[g_{m3} - g_{m2} + \frac{1}{R_3}\right] V_x$$

$$I_{out} = \left[g_{m3}(1 - g_{m1}R_1) - g_{m2} + \frac{1}{R_3} \right] V_{in}$$

$$G_m = g_{m3}(1 - g_{m1}R_1) - g_{m2} + \frac{1}{R_3} \tag{10}$$

where G_m is the overall transconductance of the LNA, g_{m1}, g_{m2}, and g_{m3} are the transconductances of transistors M_1, M_2, and M_3, respectively, and R_1 and R_3 are the local and global feedback resistors.

R_{0ut} Calculation

$$R_{out1} = \frac{R_3}{1 - \frac{1}{A_V}} \approx R_3$$

$$R_{out2} = \frac{1}{g_{m3}}$$

$$R_{out} = R_{out1} \parallel R_{out2}$$

$$R_{out} = \frac{1}{g_{m3}} \parallel R_3 \approx \frac{1}{g_{m3}}$$

Gain Calculation

,

$$A_V = G_m \cdot R_{out}$$

$$G_m = g_{m3}(1 - g_{m1}R_1) - g_{m2} + \frac{1}{R_3}$$

$$R_{out} = \frac{1}{g_{m3}} \parallel R_3 \approx \frac{1}{g_{m3}}$$

Substituting,

$$A_V = \left[g_{m3}(1 - g_{m1}R_1) - g_{m2} + \frac{1}{R_3} \right] \cdot \frac{1}{g_{m3}}$$

$$A_V = (1 - g_{m1}R_1) - \frac{g_{m2}}{g_{m3}} + \frac{1}{g_{m3}R_3} \tag{11}$$

R_{in} Calculation

$$R_{in} = R_{in1} \parallel R_{in2}$$

Now for R_{in1} ,

$$\frac{V_{in} - V_B}{R_1} = I_{in}$$

$$\frac{V_B - V_{in}}{R_1} + \frac{V_{in} - 0}{r_{ds1}} + g_{m1}V_{in} = 0$$

$$V_B = V_{in} - I_{in}R_1$$

Putting equation (3) into equation (2):

$$-I_{in} + \frac{V_{in} - I_{in}R_1}{r_{ds1}} + g_{m1}V_{in} = 0$$

$$\frac{V_{in}}{r_{ds1}} + g_{m1}V_{in} = I_{in} + \frac{I_{in}R_1}{r_{ds1}}$$

$$V_{in}\left(\frac{1}{r_{ds1}} + g_{m1}\right) = I_{in}\left(\frac{r_{ds1} + R_1}{r_{ds1}}\right)$$

Therefore,

$$R_{in1} = \frac{V_{in}}{I_{in}} = \frac{r_{ds1} + R_1}{1 + g_{m1}r_{ds1}}$$

$$R_{in1} = \frac{r_{ds1} + R_1}{g_{m1}r_{ds1} + 1} \approx \frac{1}{g_{m1}}$$

$$R_{in2} = \frac{R_3}{1 - A_V} = \frac{R_3}{\frac{g_{m3}}{g_{m2}} + g_{m1}R_1} \tag{12}$$

2.3 Comprision of Tranditional and Proposed LNA

Traditional

$$(A_v)_{\text{traditional}} = -R_F\left(g_{m1} + \frac{1}{R_S}\right) \qquad = 1 - g_{m1}R_F - \frac{g_{m2}}{g_{m3}}$$

$$(F)_{\text{traditional}} = 1 + \frac{R_S}{R_F} + \frac{\gamma/\alpha}{g_{m2}}\left(\frac{1}{R_S} + \frac{3}{R_F} + \frac{2R_S}{R_F^2}\right)$$

$$(R_{in})_{\text{traditional}} = \frac{r_{ds1} + R_F}{g_{m1}r_{ds1} + 1}$$

Propose

$$(A_V)_{\text{proposed}} = 1 - g_{m1}R_1 - \frac{g_{m2}}{g_{m3}} + \frac{1}{g_{m3}R_3}$$

$$(R_{in})_{\text{proposed}} = R_{in1} \parallel R_{in2}$$

$$(R_{out})_{\text{proposed}} = \frac{1}{g_{m3}} \parallel R_3 \approx \frac{1}{g_{m3}}$$

- The introduction of local (R_1) and global (R_3) feedback networks modifies the effective transconductance of M_2 to $\left(g_{m2} - \frac{1}{R_3}\right)$.
- This feedback structure allows the use of a larger R_1 while still maintaining proper input matching, leading to improved overall performance.
- Compared to the conventional gain expression, $|A_V|$ may initially appear reduced due to the decrease in the effective g_{m2} term.
- However, the dual-resistor feedback scheme compensates for this by enabling higher R_1, which enhances gain and noise performance without degrading input matching.
- In contrast, a single-resistor feedback LNA requires a smaller R_1 to maintain matching, limiting its achievable gain and performance.

3 Circuit Design Parameter

A. Device Dimension

Design Justification

The transistor dimensions used in this work are derived from a reference design implemented in 65 nm CMOS technology. In order to implement the same circuit in a 180 nm CMOS process, appropriate scaling of device dimensions is required. Since the minimum channel length of the target technology (180 nm) is larger than that of the reference (65 nm), the channel length of all MOSFETs is adjusted to 0.18 μm. To maintain comparable electrical behavior—particularly transconductance (g_m), current density, and overdrive voltage (V_{ov})—the transistor widths are scaled proportionally using the technology scaling factor:

$$\text{Scaling Factor} = \frac{180}{65} \approx 2.77$$

This scaling ensures that the aspect ratio (W/L), and hence the effective drive strength of the transistors, is preserved as closely as possible across technologies. Maintaining similar W/L ratios is essential for preserving key analog performance parameters such as gain, noise figure, and input matching in RF circuits like low-noise amplifiers (LNAs). It is important to note that this approach provides a first-order approximation. Due to differences in process parameters such as carrier mobility (μ), oxide capacitance (C_{ox}), and threshold voltage (V_{th}), further fine-tuning of device dimensions and bias conditions is necessary to achieve optimal performance in the 180 nm technology.

Table 1: Transistor and Component Dimensions (180 nm Technology)

Component	**W (μm)**	**L (μm)**	**nf**	**m**
M_{1p}	221.6	0.18	14	5
M_{1n}	221.6	0.18	14	5
M_2	277	0.18	13	10
M_{3n}	221.6	0.18	15	3
M_{3p}	38.8	0.18	27	1
R_1 (2.8 kΩ)	0.042	19.62	–	1
R_2 (700 Ω)	0.575	65.97	–	1
R_3 (2 kΩ)	0.6	65.97	–	2
C_2 (5 pF)	25	65.97	–	2
C_3 (1.5 pF)	2.25	65.97	–	1
C_{out} (2 pF)	30	65.97	–	1
C_{in} (2 pF)	30	65.97	–	2

Final Device Optimization

After obtaining the initial transistor dimensions through technology scaling from 65 nm to 180 nm, further optimization was performed to achieve the desired RF performance. The scaled dimensions serve only as a first-order approximation. Therefore, the transistor sizes were fine-tuned through iterative simulations by analyzing key performance metrics such as Noise Figure (NF) and S-parameters (S_{11}, S_{21}, and S_{22}). These results were then compared with the reference design to ensure comparable performance.

In particular, the transconductance (g_m) of critical transistors was carefully adjusted by modifying their widths. For instance, the output stage transistor (M_3) was tuned to achieve proper output impedance matching. By controlling g_{m3}, the output reflection coefficient (S_{22}) was improved, ensuring better power transfer and stability.
Similarly, the input transistors were optimized to achieve good input matching (S_{11}) and low noise figure. This iterative design approach ensures that the final dimensions are not only scaled but also performance-optimized for the 180 nm technology.
The final optimized device dimensions used in this work are summarized in Table **??**.

Components	**W (μm)**	**L (μm)**	**nf (μm)**	**m=1**
M_{1p}	68	0.18	13.8	5
M_{1n}	68	0.18	13.8	5
M_2	130	0.18	13	10
M_{3n}	46.14	0.18	15.38	3
M_{3p}	26.9	0.18	26.9	1
R_1 (2.8kΩ)	0.042	19.62	N/A	1
R_2 (700Ω)	0.575	65.97	–	1
R_3 (2kΩ)	0.6	65.965	–	2
C_2 (5pF)	25	65.965	–	2
C_3 (1.50pF)	2.25	65.965	–	1
C_{out} (2pF)	30	65.965	–	1
C_{in} (2pF)	30	65.965	–	2

Table 2: Component Dimensions Table

4 Cadence Simulation Setup

4.1 Noise Analysis

Thermal Noise

Thermal noise (also called white noise) in a MOSFET is generated due to the random motion of charge carriers in the channel. It is present in both NMOS and PMOS transistors and is independent of frequency (i.e., flat over frequency). The drain current thermal noise power spectral density is given by:

$$\overline{i_n^2} = 4k_B T \gamma g_m$$

where k_B is Boltzmann's constant, T is the absolute temperature, γ is the channel thermal noise coefficient (typically $\approx 2/3$ for long-channel devices and higher for short-channel), and g_m is the transconductance of the MOSFET. **Key Points:**

- Thermal noise is frequency independent (white noise).
- It is directly proportional to g_m and temperature.
- Present in both NMOS and PMOS devices with similar behavior.

Flicker Noise (1/f Noise)

Flicker noise, also known as $1/f$ noise, is caused by charge trapping and detrapping at the oxide-semiconductor interface. It dominates at low frequencies and decreases as frequency increases.

The drain current flicker noise power spectral density is given by:

$$\overline{i_n^2} = \frac{K}{C_{ox}^2 WL} \cdot \frac{g_m^2}{f}$$

where K is a process-dependent constant, C_{ox} is the oxide capacitance per unit area, W and L are the width and length of the MOSFET, g_m is the transconductance, and f is the frequency.

Key Points:

- Flicker noise is inversely proportional to frequency ($\propto 1/f$).
- Dominates at low frequencies.
- Reduced by increasing device area ($W \times L$).
- PMOS devices typically exhibit lower flicker noise than NMOS due to lower trap density.

Frequency Dependence of Noise

The total noise in a MOSFET is the combination of thermal and flicker noise:

- At low frequencies: Flicker noise dominates.
- At high frequencies: Thermal noise dominates.
- Transition occurs at the corner frequency (f_c), where both noise contributions are equal.

Relevance in Cadence Noise Analysis

In Cadence simulations:

- Noise analysis computes output and input-referred noise across frequency.
- At low frequency, the noise curve shows a $1/f$ slope due to flicker noise.
- At high frequency, the noise becomes flat due to thermal noise.
- NMOS devices generally contribute higher flicker noise compared to PMOS.

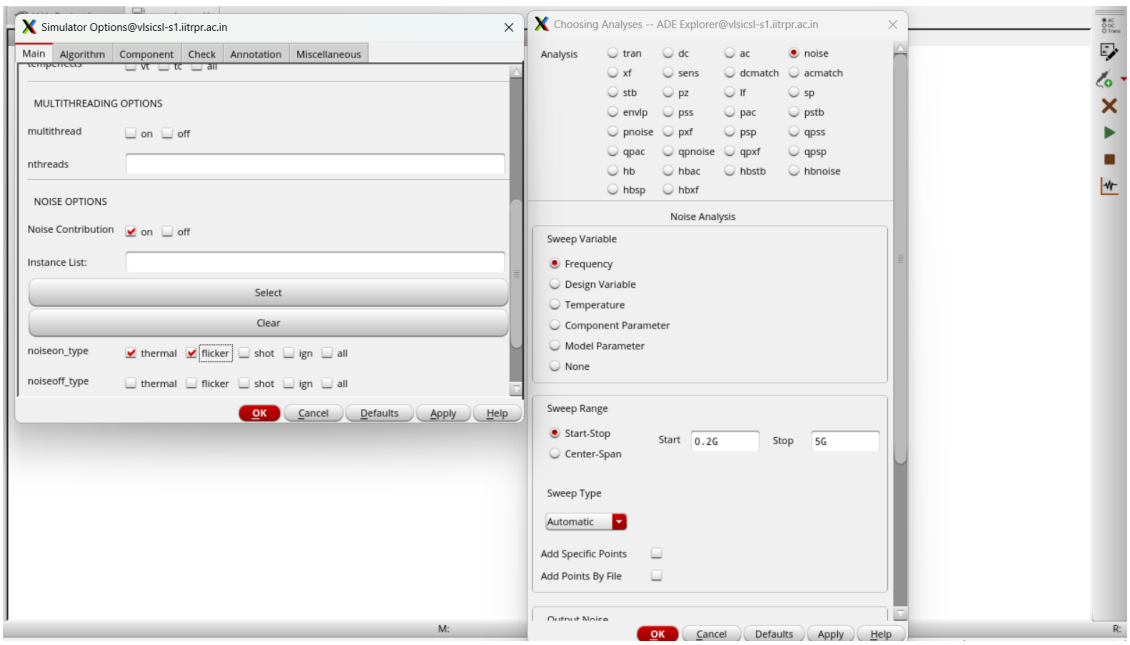


Figure 4: Noise Setup

4.2 S-Parameter Analysis

S-parameters (Scattering parameters) are widely used to characterize high-frequency circuits such as RF amplifiers. They describe how RF signals behave at the input and output ports of a network in terms of reflection and transmission. For a two-port network, the S-parameters are defined as:

- S_{11}: Input reflection coefficient
- S_{21}: Forward gain (transmission from input to output)
- S_{12}: Reverse isolation (transmission from output to input)
- S_{22}: Output reflection coefficient

Significance of S_{11}

S_{11} represents the input reflection coefficient of the amplifier. It indicates how much of the input signal is reflected back due to impedance mismatch.

- A lower value of $|S_{11}|$ (typically < -10 dB) indicates good input matching.
- Good input matching ensures maximum power transfer from the source to the amplifier.
- Poor S_{11} results in signal reflection and reduced efficiency.

Significance of S_{12}

S_{21} represents the forward transmission coefficient of the amplifier, indicating how much of the input signal is transmitted to the output.

- It directly represents the gain of the amplifier.
- A higher value of $|S_{21}|$ indicates better amplification capability.
- It is usually expressed in decibels as:

$$S_{21}(\text{dB}) = 20 \log |S_{21}|$$

- A flat S_{21} over the operating frequency range ensures consistent gain.
- It is a key parameter for evaluating the performance of RF amplifiers such as LNAs.

Relevance in Cadence Simulation

In Cadence SP analysis:

- S_{11} is used to verify input impedance matching across frequency.
- S_{21} indicates the gain of the amplifier.
- S_{12} helps evaluate reverse isolation and stability.
- S_{22} ensures proper output matching.

A well-designed LNA should exhibit low S_{11}, high S_{21}, low S_{12}, and low S_{22} over the desired frequency range.

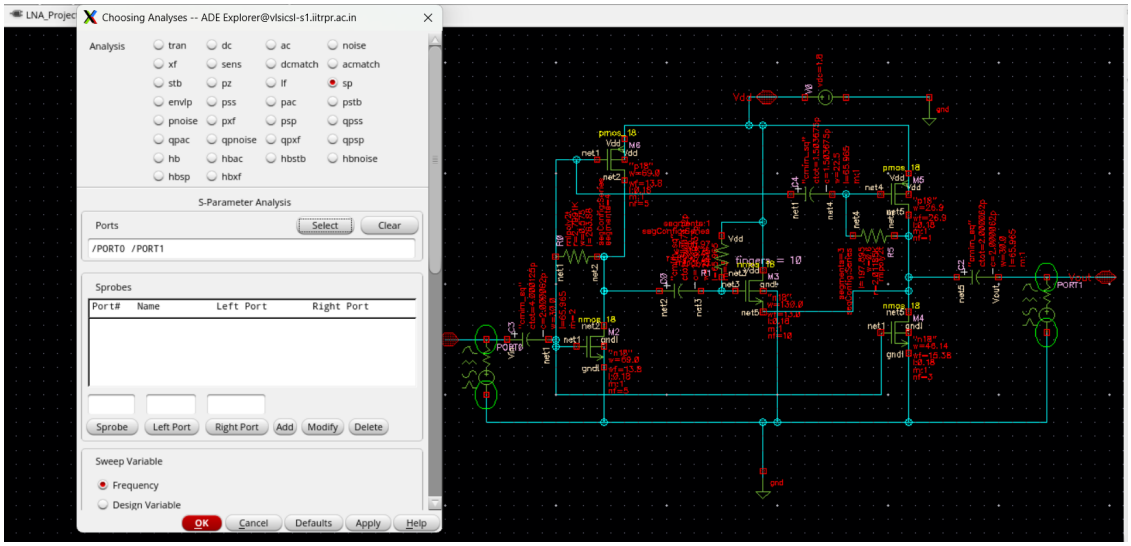


Figure 5: SP Setup

5 Simulation Results

5.1 Pre Layout Simulation Results

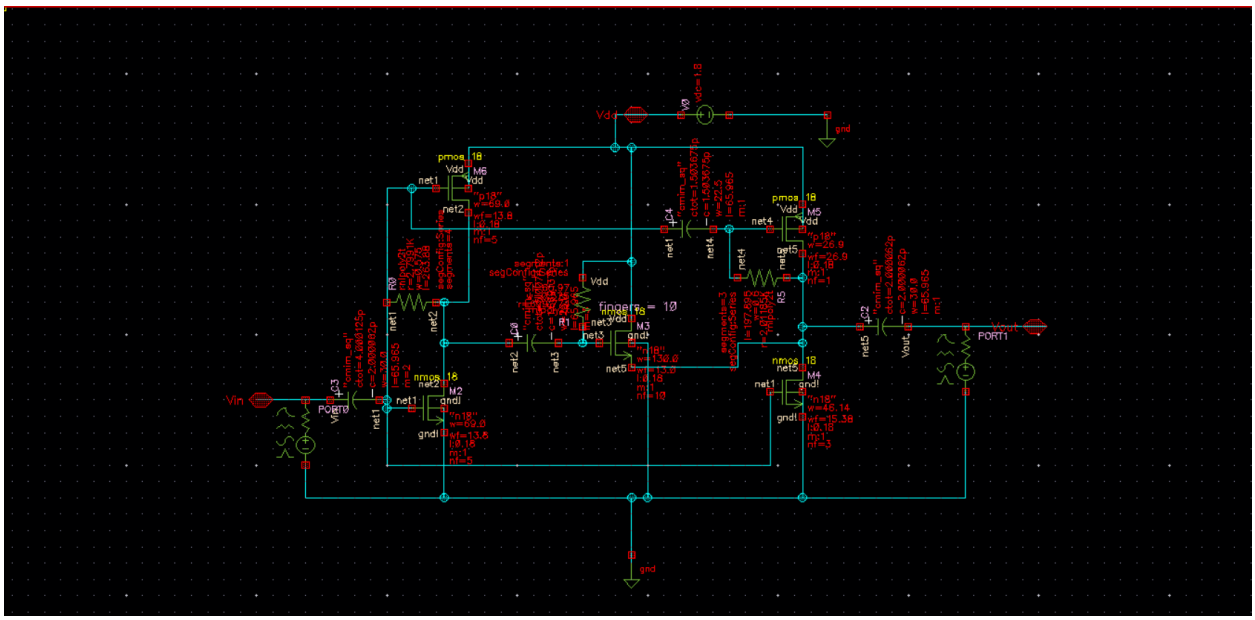


Figure 6: Schematic of LNA

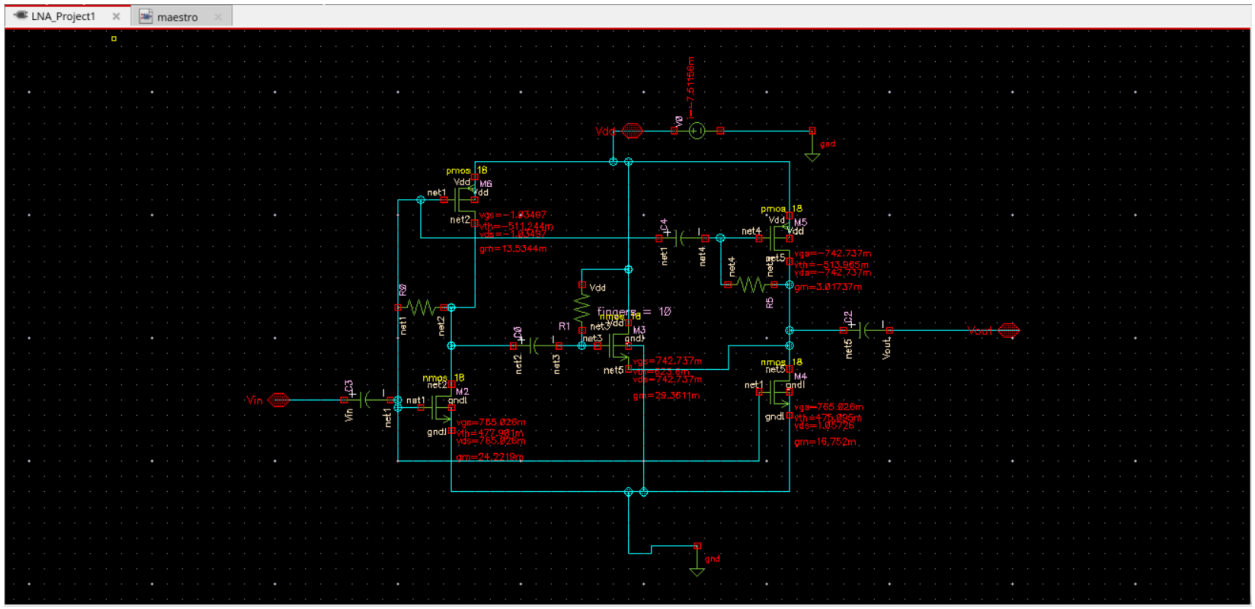


Figure 7: DC Annotation

fParameters

MOSFET	g_m (mS)
M_1	24.22
M_{1p}	13.53
M_2	29.36
M_{B0}	16.75
M_{3p}	3.01

Table 3: g_m Matrix

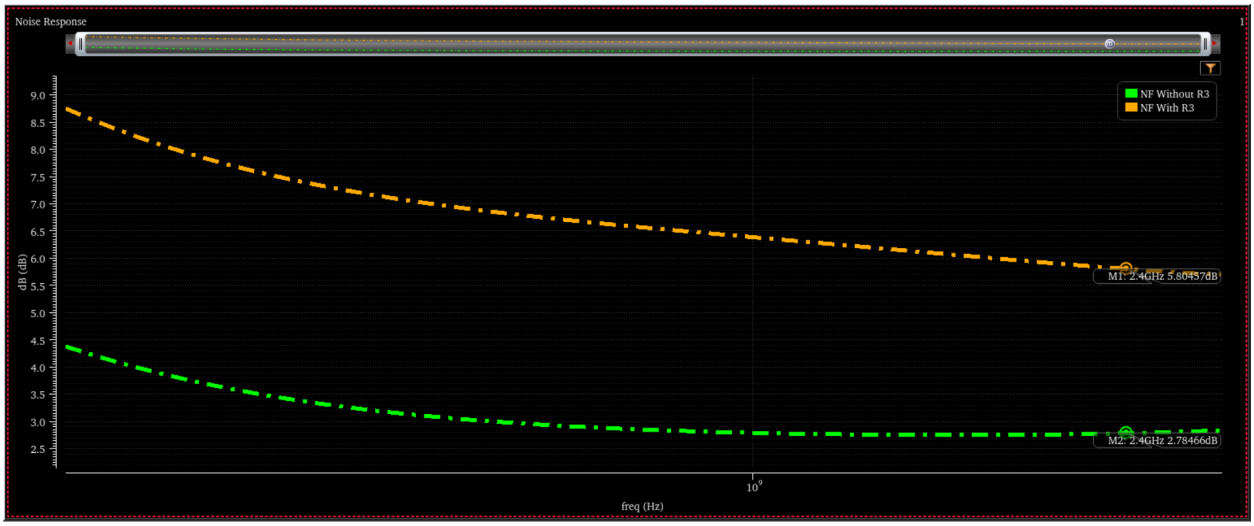


Figure 8: Noise Figure with or Without R_3

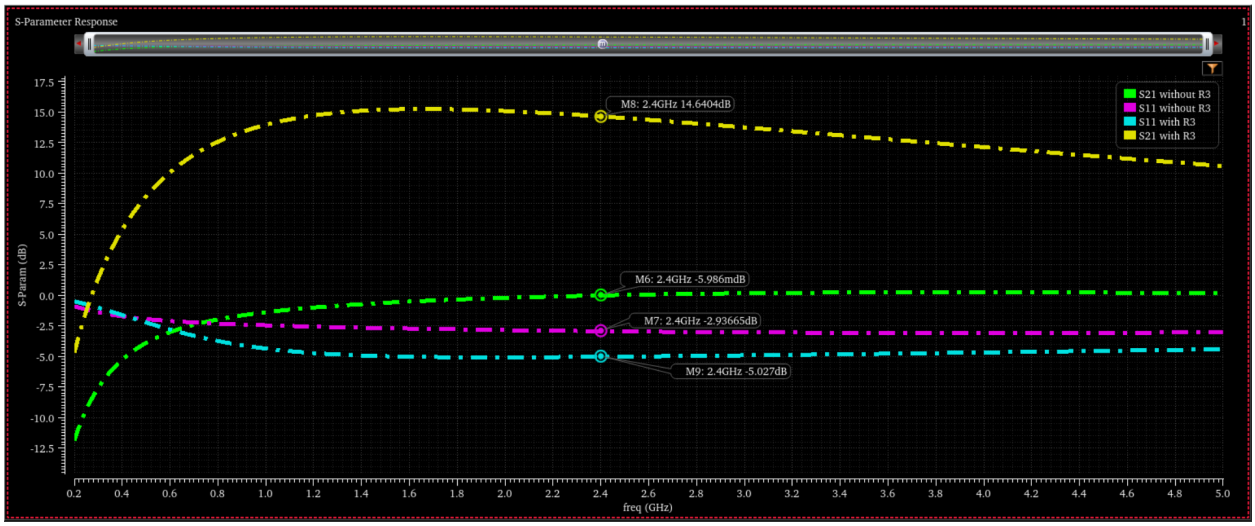


Figure 9: S-Parameter with or Without R_3

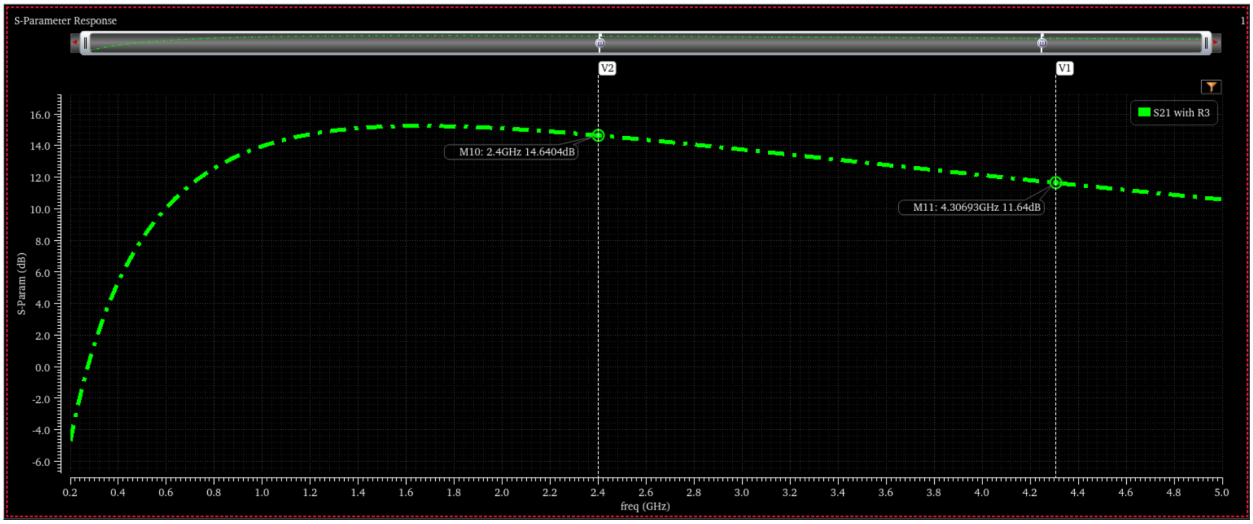


Figure 10: S-Parameter(21) with R_3

$$BW = f_H - f_L = 4.30\,\text{GHz} - 2.40\,\text{GHz} = 1.90\,\text{GHz}$$

5.2 Layout

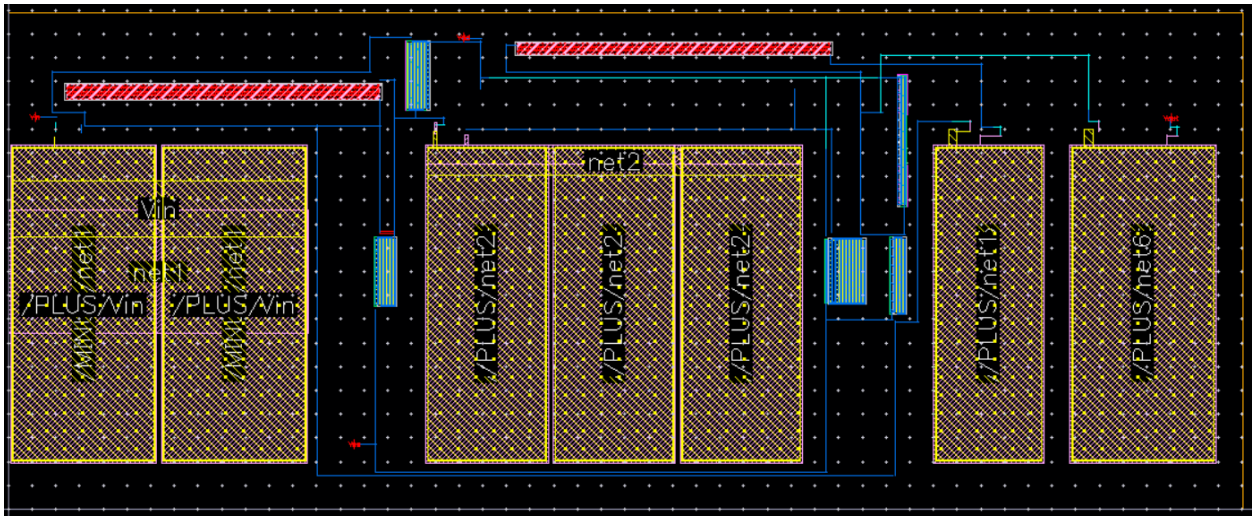


Figure 11: Layout of LNA

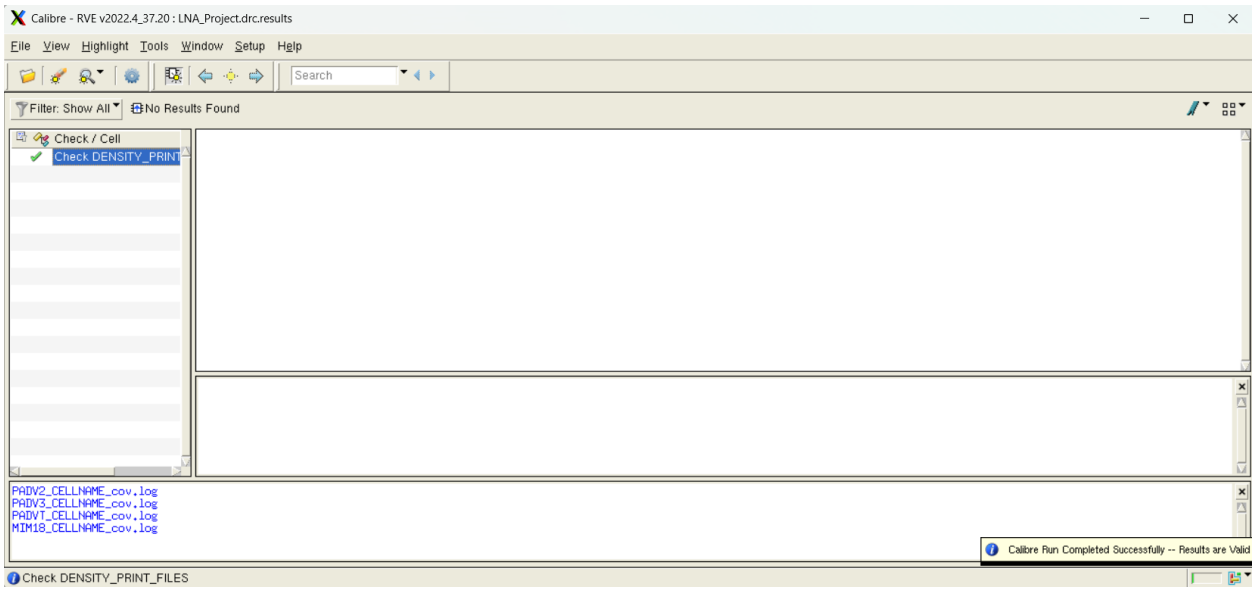


Figure 12: DRC Status

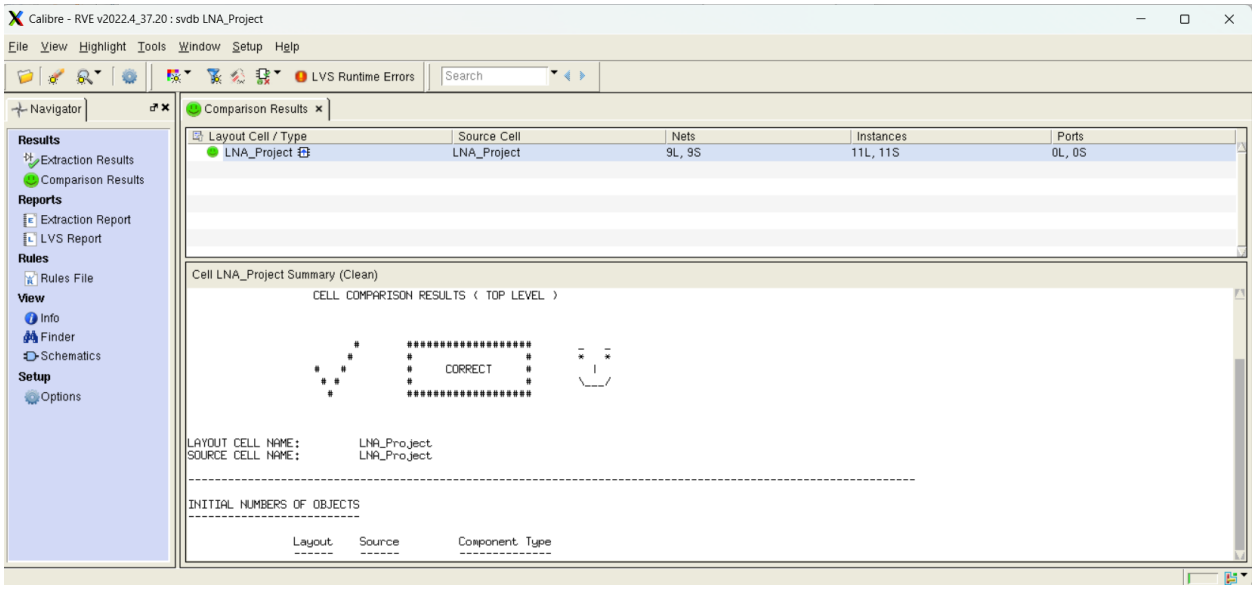


Figure 13: LVS Status

5.3 Post Layout Simulation Results

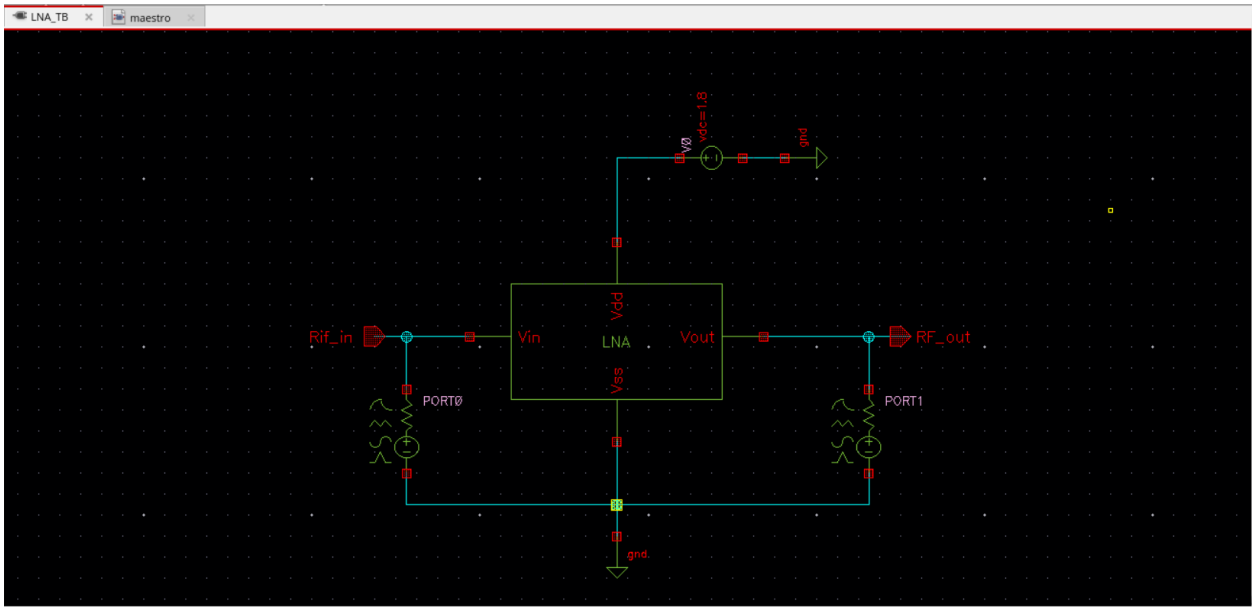


Figure 14: Test Banch for Post Layout

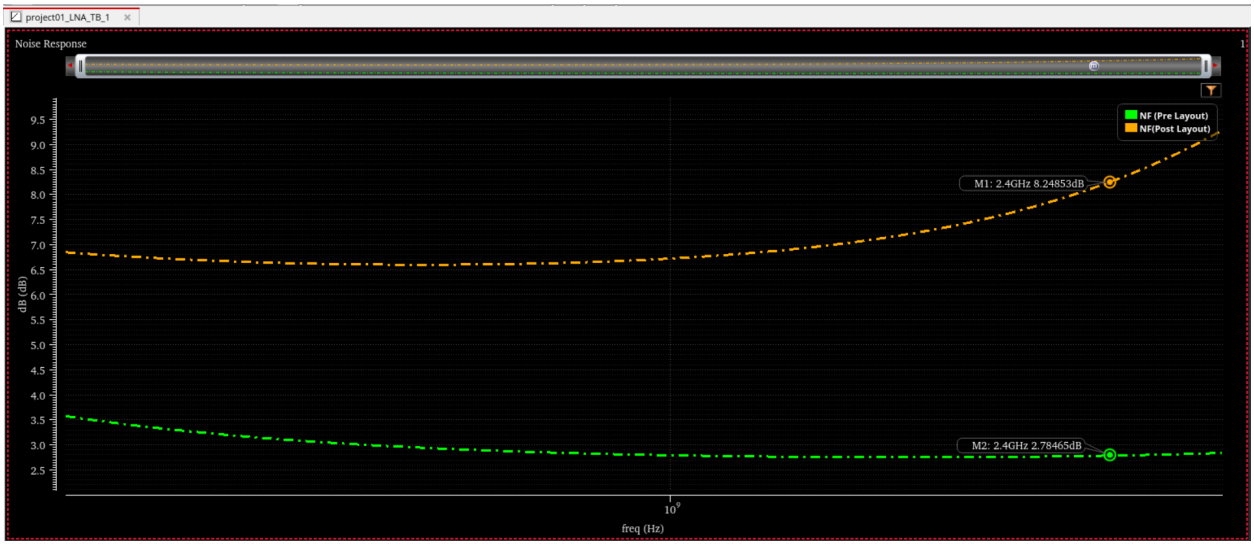


Figure 15: Noise Figure (Pre Vs Post Layout)

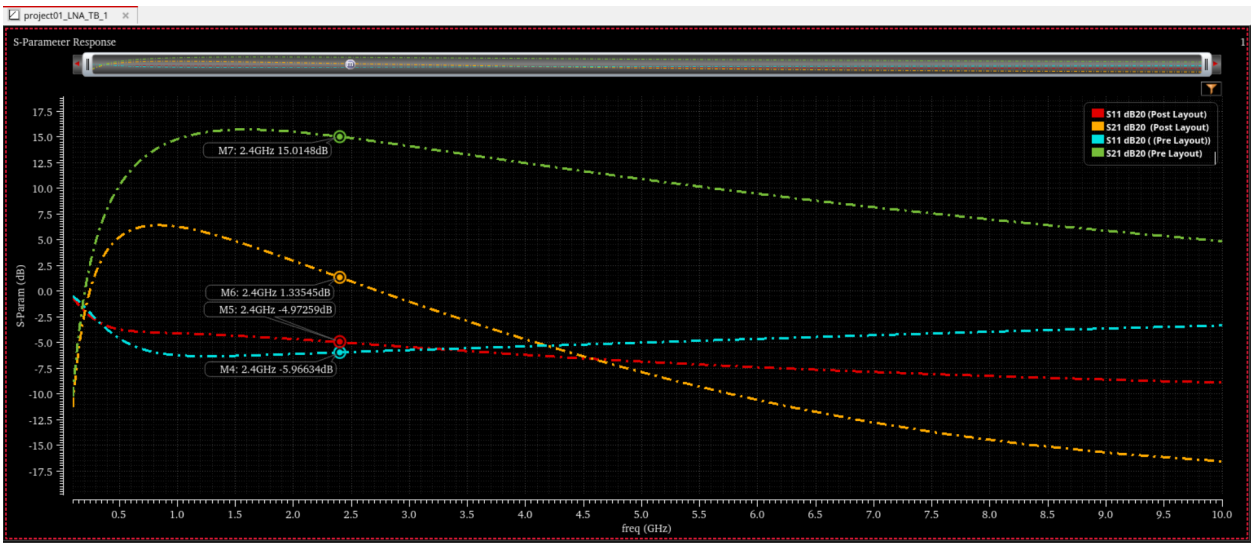


Figure 16: S-Parameter (Pre Vs Post Layout)

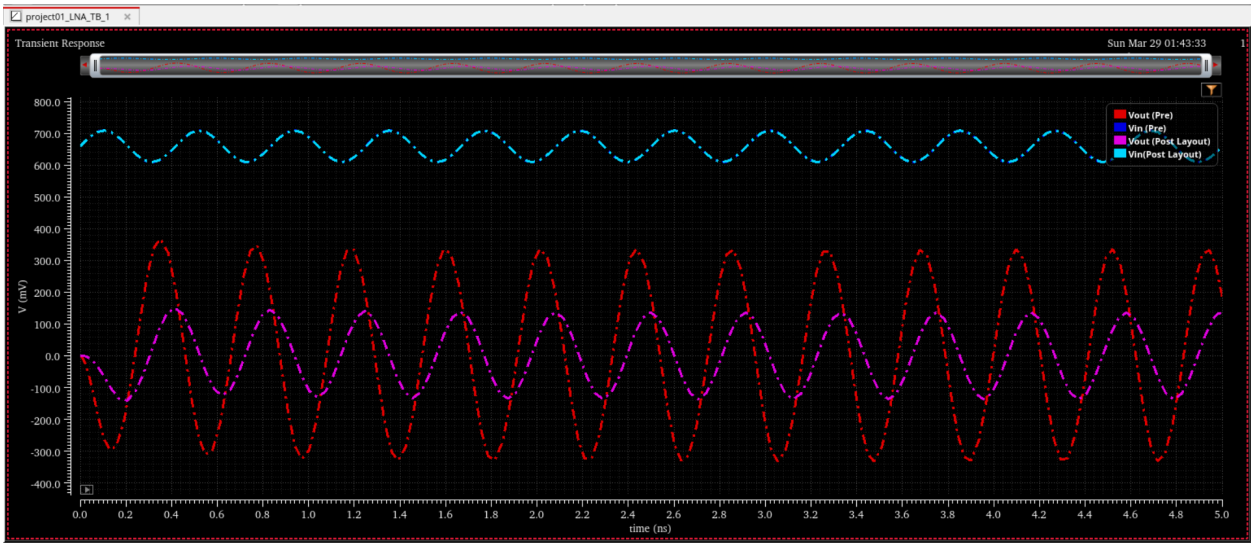


Figure 17: Transient Response (Pre Vs Post Layout)

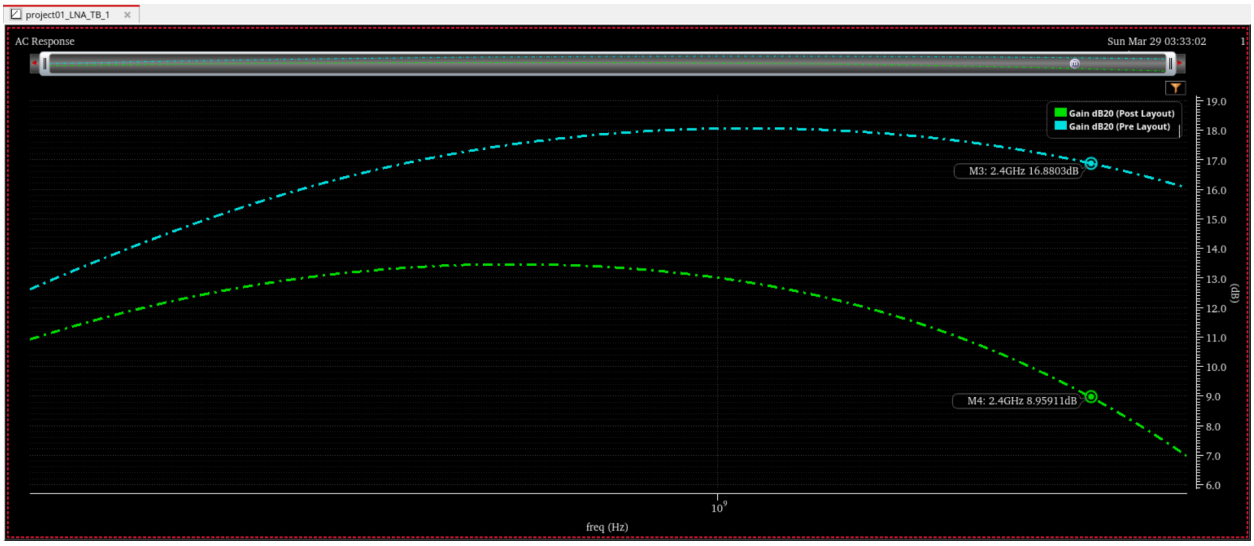


Figure 18: AC Response (Pre Vs Post Layout)

5.4 Padframe

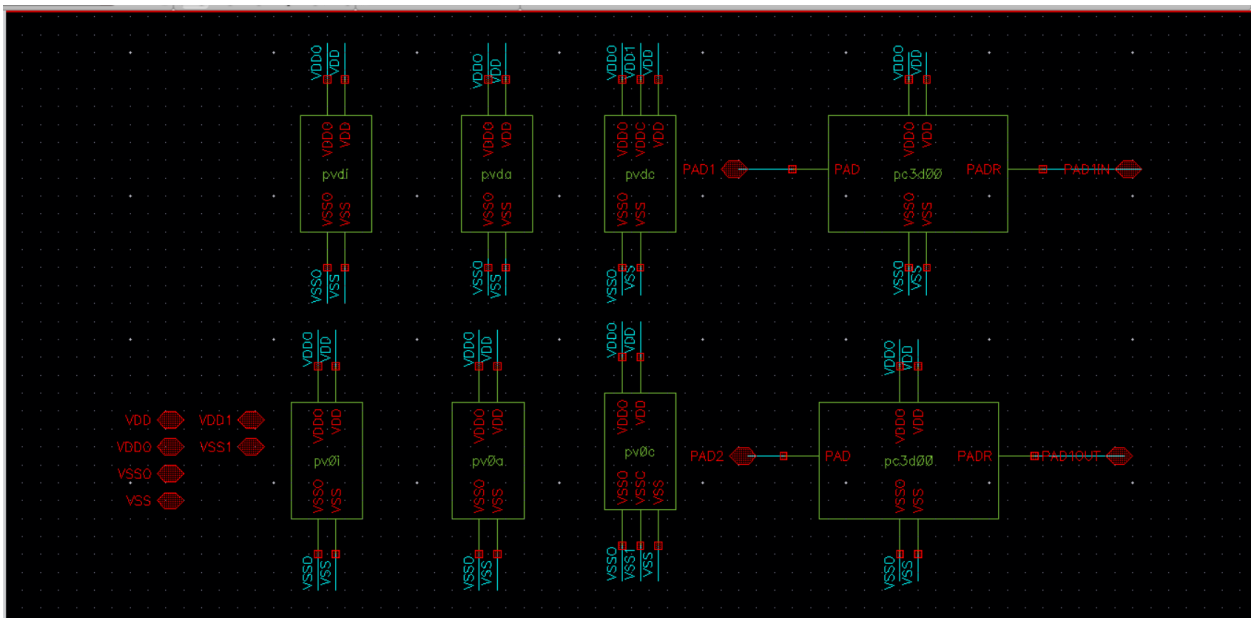


Figure 19: Schematics of Pads

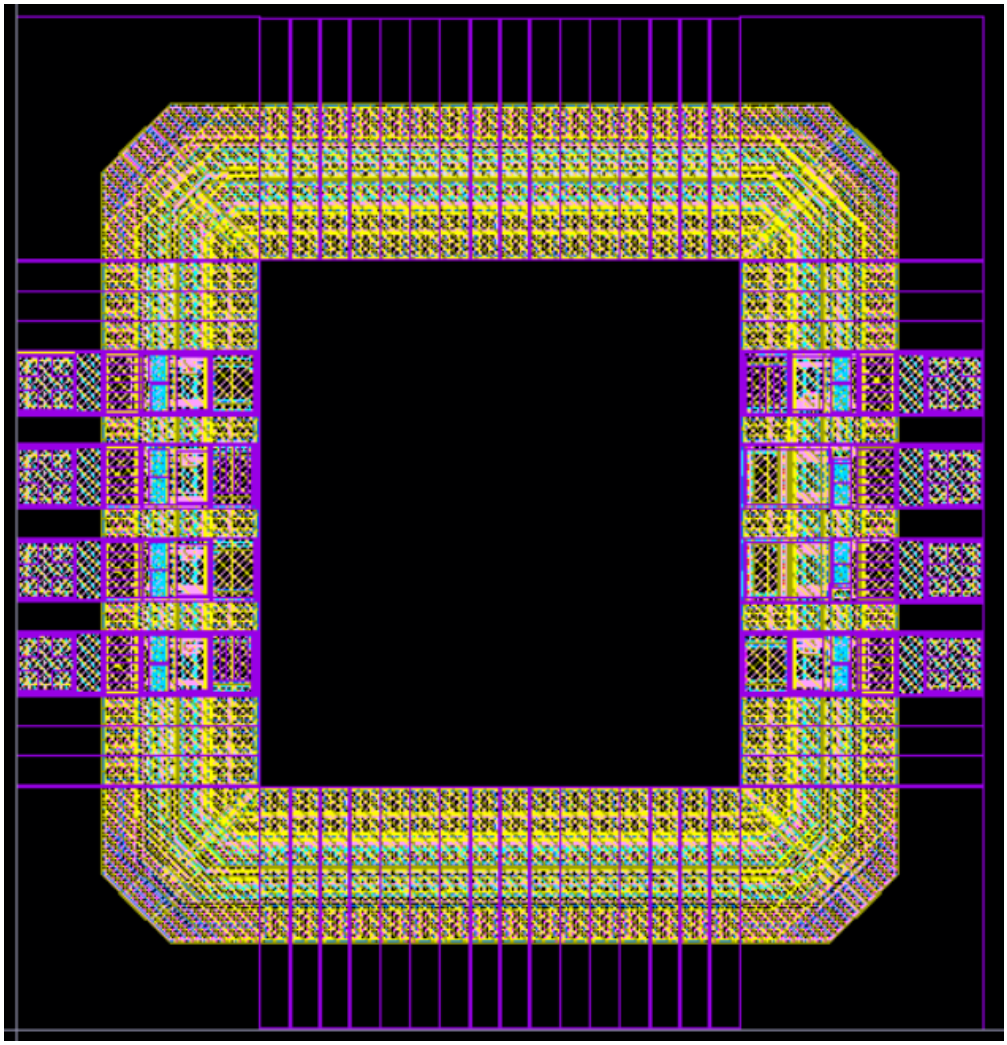


Figure 20: Padframe Layout

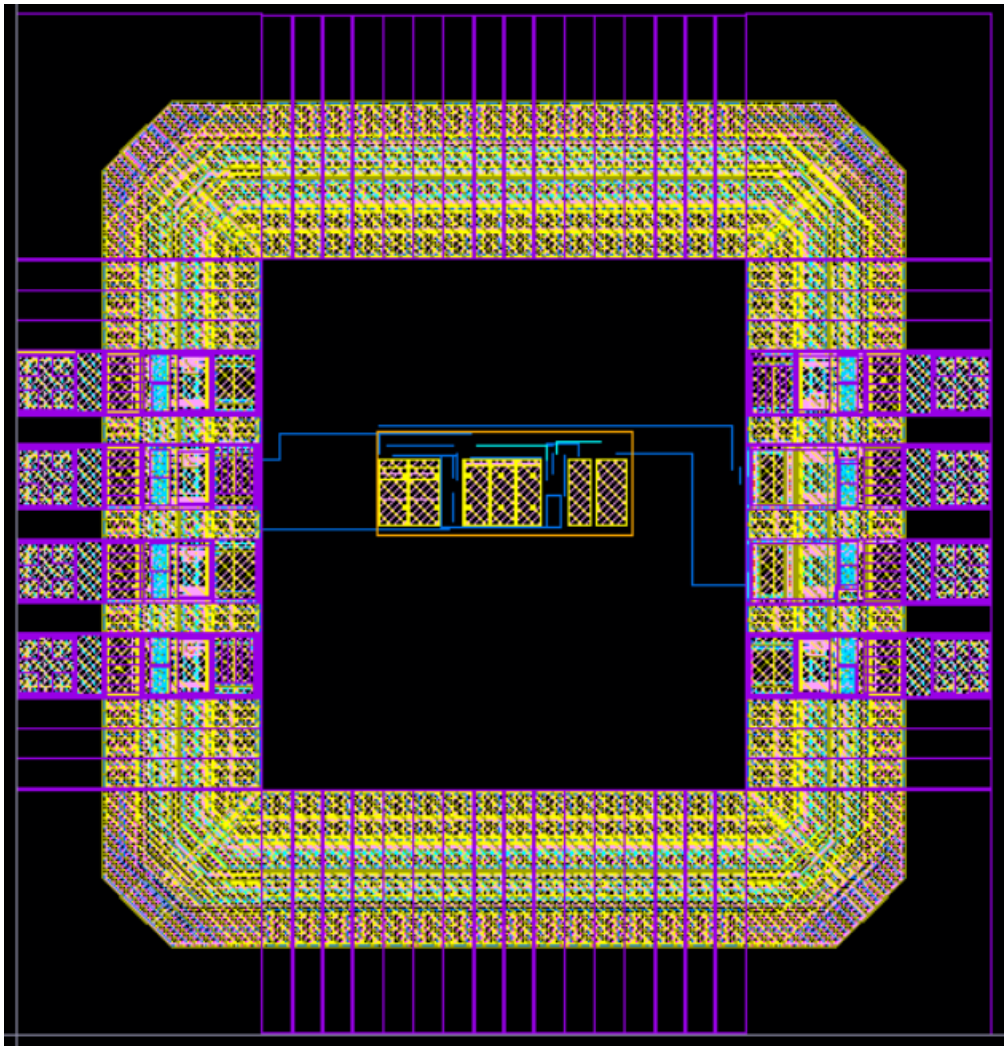


Figure 21: Padframe Layout(Intigration with LNA)

6 Impedance Matching

6.1 Output Impedance

$$(R_{out})_{\text{proposed}} = \frac{1}{g_{m3}} \parallel R_3 \approx \frac{1}{g_{m3}}$$

$$g_{m3} = g_{m3n} + g_{m3p} = 19.76\,\text{mS}, \quad R_3 = 2\,\text{k}\Omega$$

$$(R_{out})_{\text{proposed}} = \frac{1}{g_{m3}} \parallel R_3 = \frac{1}{19.76 \times 10^{-3}} \parallel 2000$$

$$(R_{out})_{\text{proposed}} \approx 50.6\,\Omega \parallel 2000\,\Omega \approx 49.4\,\Omega \approx 50\,\Omega$$

7 Performance Summary

Parameter	Value
Technology Node	180 nm
Power	13.518 mW
V_{DD}	1.8 V
NF (with R_3)	5.80 dB
NF (without R_3)	2.78 dB
Gain (with R_3)	14.64 dB
Gain (without R_3)	-5.98 dB
Current	7.51 mA
Bandwidth	1.90 GHz

Table 4: Performance Table

8 Conclusion

In this work, an inductor-less low-noise amplifier (LNA) based on noise-canceling and dual-resistor feedback techniques has been successfully designed and analyzed. The proposed architecture effectively addresses the fundamental trade-offs between gain, noise figure, and input matching that are typically observed in conventional LNA designs. Theoretical analysis of key parameters such as transconductance, input impedance, output impedance, and voltage gain has been carried out using small-signal models. The results demonstrate that the dual-resistor feedback structure enables improved performance by allowing higher gain and reduced noise while maintaining acceptable impedance matching. Simulation results obtained from Cadence validate the analytical findings. The S-parameter analysis confirms good input and output matching, along with satisfactory gain and isolation across the desired frequency range. Noise analysis further shows that the implementation of the noise-canceling technique significantly reduces the overall noise figure of the amplifier. Although minor deviations between theoretical and simulated results are observed due to non-ideal effects and parasitics, the overall performance meets the design objectives. The proposed LNA achieves a good balance between gain, bandwidth, noise performance, and power consumption. Hence, the designed LNA is suitable for modern broadband RF applications, particularly in sub-6 GHz communication systems, where compact area, low power, and high performance are essential.

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